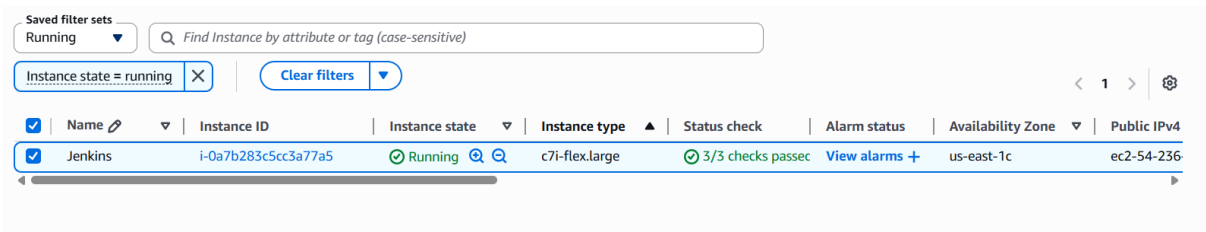


JENKINS

1. Install Jenkins and run Jenkins on port number 8081.

- Created a new EC2 Instance for Jenkins



- Connected the EC2 and install Jenkins

```
complete!
[ec2-user@ip-172-31-21-129 ~]$ sudo yum install jenkins
Last metadata expiration check: 0:00:29 ago on Mon May 11 10:52:53 2026.
Dependencies resolved.
=====
Package                        Architecture      Version           Repository        Size
=====
Installing:
jenkins                       noarch            2.555.1-1        jenkins           95 M
=====
Transaction Summary
=====
Install 1 Package
```

- Java installed

```
complete!
[ec2-user@ip-172-31-21-129 ~]$ sudo yum install java-21-amazon-corretto -y
Last metadata expiration check: 0:00:15 ago on Mon May 11 10:52:53 2026.
Dependencies resolved.
=====
Package                        Arch      Version           Repository        Size
=====
Installing:
java-21-amazon-corretto       x86_64    1:21.0.11+10-1.amzn2023.1  amazonlinux       218 k
Installing dependencies:
alsa-lib                      x86_64    1.2.7.2-1.amzn2023.0.3  amazonlinux        503 k
cairo                          x86_64    1.18.0-4.amzn2023.0.3  amazonlinux        717 k
dejavu-sans-fonts              noarch    2.37-16.amzn2023.0.2  amazonlinux        1.3 M
dejavu-sans-mono-fonts         noarch    2.37-16.amzn2023.0.2  amazonlinux        467 k
dejavu-serif-fonts             noarch    2.37-16.amzn2023.0.2  amazonlinux        1.0 M
fontconfig                     x86_64    2.13.94-2.amzn2023.0.2  amazonlinux        273 k
fonts-filesystem               noarch    1:2.0.5-12.amzn2023.0.2  amazonlinux        9.5 k
```

- Start Jenkins and check the status is active or not

```
[ec2-user@ip-172-31-21-129 ~]$ sudo systemctl daemon-reload
[ec2-user@ip-172-31-21-129 ~]$ sudo systemctl restart jenkins
[ec2-user@ip-172-31-21-129 ~]$ sudo systemctl start jenkins
[ec2-user@ip-172-31-21-129 ~]$ sudo systemctl status jenkins
jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; disabled; preset: disabled)
   Active: active (running) since Mon 2026-05-11 10:56:55 UTC; 19s ago
     Main PID: 26665 (java)
       Tasks: 50 (limit: 4523)
      Memory: 581.4M
         CPU: 10.880s
        CGroup: /system.slice/jenkins.service
                └─26665 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/v

May 11 10:56:53 ip-172-31-21-129.ec2.internal jenkins[26665]: [LF]> This may also be found at: /var/lib/j
May 11 10:56:53 ip-172-31-21-129.ec2.internal jenkins[26665]: [LF]>
May 11 10:56:53 ip-172-31-21-129.ec2.internal jenkins[26665]: [LF]> *****
May 11 10:56:53 ip-172-31-21-129.ec2.internal jenkins[26665]: [LF]> *****
May 11 10:56:53 ip-172-31-21-129.ec2.internal jenkins[26665]: [LF]> *****
May 11 10:56:55 ip-172-31-21-129.ec2.internal jenkins[26665]: 2026-05-11 10:56:55.897+0000 [id=40]
May 11 10:56:55 ip-172-31-21-129.ec2.internal jenkins[26665]: 2026-05-11 10:56:55.909+0000 [id=30]
May 11 10:56:55 ip-172-31-21-129.ec2.internal systemd[1]: Started jenkins.service - Jenkins Continuous In
May 11 10:56:56 ip-172-31-21-129.ec2.internal jenkins[26665]: 2026-05-11 10:56:56.182+0000 [id=56]
May 11 10:56:56 ip-172-31-21-129.ec2.internal jenkins[26665]: 2026-05-11 10:56:56.184+0000 [id=56]
```

- Change the port number from 8080 to 8081

```
# Port to listen on for HTTP requests. Set to -1 to disable.
# To be able to listen on privileged ports (port numbers less than 1024)
# add the CAP_NET_BIND_SERVICE capability to the AmbientCapabilities
# directive below.
Environment="JENKINS_PORT=8080"
```

```
# add the CAP_NET_BIND_SERVICE capability to the AmbientCapabilities
# directive below.
Environment="JENKINS_PORT=8081"

# IP address to listen on for HTTPS requests. Default is disabled
#Environment="JENKINS_HTTPS_LISTEN_ADDRESS="
```

- Check the browser port number 8081



- Installed suggest plugin and created First Admin user

Getting Started

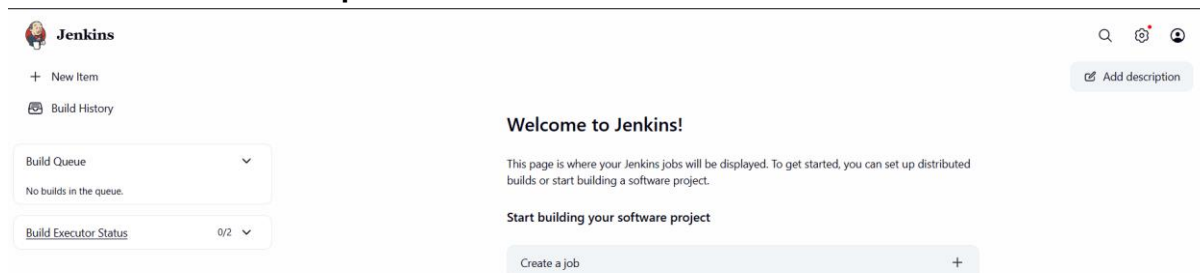
Create First Admin User

Username

Password

Confirm password

- Done with Jenkins setup



2. Secure Jenkins server

- Open plugins and in that available plugins and install Role based strategy

The screenshot shows the Jenkins 'Manage Jenkins' > 'Plugins' page. A search bar at the top contains the word 'Security'. On the left sidebar, 'Available plugins' is selected. The main table lists two plugins:

Install	Name ↓	Released	Health
☐	PAM Authentication 1.12 Security Adds Unix Pluggable Authentication Module (PAM) support to Jenkins	1 yr 2 mo ago	90
☒	Role-based Authorization Strategy 848.va_a_ea_673cf0b_c Security Authentication and User Management Enables user authorization using a Role-Based strategy. Roles can be defined globally or for particular jobs or nodes selected by regular expressions.	5 mo 9 days ago	97

- Installed Role based Strategy

The screenshot shows the Jenkins 'Manage Jenkins' > 'Plugins' page with 'Installed plugins' selected in the sidebar. A search bar contains the word 'role'. The table shows one installed plugin:

Name ↓	Health	Enabled
Role-based Authorization Strategy 848.va_a_ea_673cf0b_c Enables user authorization using a Role-Based strategy. Roles can be defined globally or for particular jobs or nodes selected by regular expressions. Report an issue with this plugin	97	☒

- In security check authorization select Role based Strategy and save it.

The screenshot shows the Jenkins 'Manage Jenkins' > 'Security' configuration page. It includes the following settings:

- ☐ Disable "Keep me signed in" ?
- Security Realm: Jenkins' own user database
- ☐ Allow users to sign up ?
- Authorization: Role-Based Strategy

3. Create users called DevOps, Testing in Jenkins with Limited access.

- Open manage Jenkins and click on users and created new

The screenshot shows the Jenkins 'Manage Jenkins' > 'Jenkins' own user database' page. It features a '+ Create User' button and a table of users:

User ID ↓	Name
Arunsai	Kathi Arunsai

- **Created new user with name DevOps**

je Jenkins  / Jenkins' own user database / Create User

Create User

Username

DevOps

Password

.....

Confirm password

.....

Full name

DevOps

E-mail address

devops@gmail.com

- **Created another user with name Testing**

Create User

Username

Testing

Password

.....

Confirm password

.....

Full name

Testing

E-mail address

testing@gmail.com

Users 3

+ Create User

These users can log into Jenkins. This is a sub set of [this list](#), which also contains auto-created users who really just made some commits on some projects and have no direct Jenkins access.

User ID ↓	Name		
Arunsai	Kathi Arunsai		
DevOps	DevOps		
Testing	Testing		

- In Manage roles added DevOps, Testing and Admin Permissions

Manage Roles

Global roles

Role		Overall	Credentials				Agent					Job					Run	View			SCM	Ma												
		Administer	Manage	Read	Create	Delete	ManageDomains	Update	View	Build	Configure	Connect	Create	Delete	Disconnect	Provision	Build	Cancel	Configure	Create	Delete	Discover	Move	Read	Workspace	Delete	Replay	Update	Configure	Create	Delete	Read	Tag	HealthCheck
admin																																		
DevOps																																		
Testing																																		

- In Assign Role We can check the attached permissions

manage domains / manage and assign roles / assign roles

Assign Roles

es

OS

Global roles

User/Group	DevOps	Testing	admin	
Anonymous	☐	☐	☐	
Authenticated Users	☐	☐	☐	
Kathi Arunsai	☐	☐		
DevOps		☐	☐	
Testing	☐		☐	

Add User

Add Group

4. Configure labels and restrict the jobs to execute based on label only.

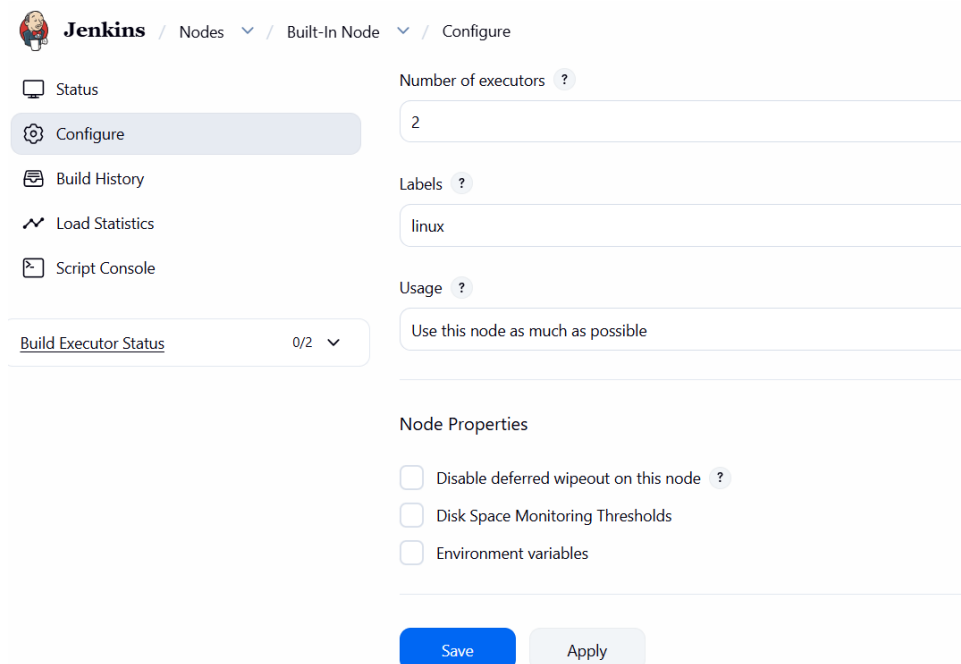
- Open Manage Jenkins and open Nodes and open Built in Node



The screenshot shows the Jenkins 'Nodes' page. At the top, there's a breadcrumb: 'Jenkins / Manage Jenkins / Nodes'. Below this, a table lists the nodes. The first node is 'Built-In Node', which is a Linux (amd64) architecture, in sync, with 5.64 GiB free disk space, 0 B free swap space, and 1.86 GiB free temp space. It has a response time of 0ms. Below the table, there are tabs for 'S', 'M', and 'L', with 'S' selected. A legend is visible at the bottom right.

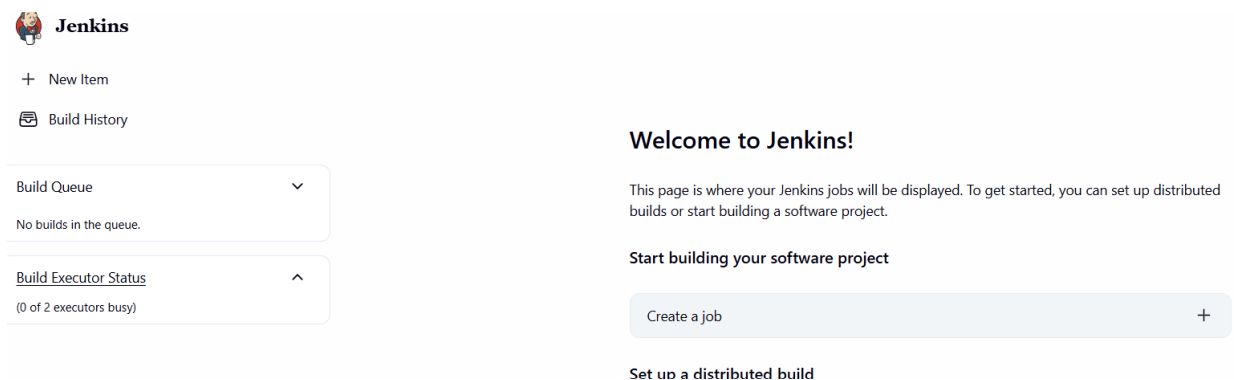
S	Name	Architecture	Clock Difference	Free Disk Space	Free Swap Space	Free Temp Space	Response Time
	Built-In Node	Linux (amd64)	In sync	5.64 GiB	0 B	1.86 GiB	0ms
	last checked	43 min	43 min	43 min	43 min	43 min	43 min

- Created a label name as Linux



The screenshot shows the 'Configure' page for the 'Built-In Node'. The left sidebar has tabs for 'Status', 'Configure', 'Build History', 'Load Statistics', and 'Script Console'. The 'Configure' tab is active. The main content area has sections for 'Number of executors' (set to 2), 'Labels' (set to 'linux'), 'Usage' (set to 'Use this node as much as possible'), and 'Node Properties' (with checkboxes for 'Disable deferred wipeout on this node', 'Disk Space Monitoring Thresholds', and 'Environment variables'). At the bottom, there are 'Save' and 'Apply' buttons.

- Now created New Item or create a job in Jenkins



The screenshot shows the Jenkins main page. On the left, there's a sidebar with 'New Item' and 'Build History' buttons. Below these, there's a 'Build Queue' section showing 'No builds in the queue.' and a 'Build Executor Status' section showing '(0 of 2 executors busy)'. On the right, there's a 'Welcome to Jenkins!' message, followed by 'Start building your software project' and a 'Create a job' button. At the bottom, there's a 'Set up a distributed build' link.

- **Now given name and choose Freestyle Project**

Jenkins / All / New Item

New Item

Enter an item name

Linux-Label-Job

Select an item type

 Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.

 Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

- **Default we can see Restrict this project run we can add label name**

s

☐ This project is parameterised ?

☐ Throttle builds ?

☐ Execute concurrent builds if necessary ?

☒ Restrict where this project can be run ?

Label Expression ?

linux

Label linux matches 1 node. Permissions or other restrictions provided by plugins may further reduce that list.

Advanced ▾

- **In Build steps select Execute Shell and given command**

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

 Execute shell ?

Command

See [the list of available environment variables](#)

```
echo "Job running on linux labeled node"
```

- **Save and run the build and check the status**

 Status

 Changes

 Workspace

 Build Now

 Configure

 Delete Project

 Rename

Build Time Trend

S	Build 1	Time Since	Duration	
	#1	11 sec	34 ms	

Icon: S M L

mins

- Checked the console running successful

Console

```
Started by user Kathi Arunsai
Running as SYSTEM
Building in workspace /var/lib/jenkins/workspace/Linux-Label-Job
[Linux-Label-Job] $ /bin/sh -xe /tmp/jenkins1553241742735484889.sh
+ echo 'Job running on linux labeled node'
Job running on linux labeled node
Finished: SUCCESS
```

5. Create Three sample jobs using the below URL.

Source Code: https://github.com/betawins/Techie_horizon_Login_app.git

- Create a new item as Github-01 in Jenkins

New Item

Enter an item name

Github-01

Select an item type



Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

- Install Git in gitbash

```
[ec2-user@ip-172-31-21-129 ~]$ sudo yum install git -y
Last metadata expiration check: 5:04:45 ago on Mon May 11 10:52:53 2026.
Package git-2.50.1-1.amzn2023.0.1.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-21-129 ~]$
```

- Select Source code management and in that select git and paste url



General



Source Code Management



Triggers



Environment



Build Steps



Post-build Actions



None



Git ?

Repositories ?

Repository URL ?

https://github.com/betawins/Techie_horizon_Login_app.git

- **In Build steps select the Execute shell and write command**

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute shell ?

Command

[See the list of available environment variables](#)

```
echo "Github-01 Job Success"
```

- **Run the Build Now and output correct**

- Status
- Changes
- Workspace
- Build Now
- Configure
- Delete Project
- Rename

Build Time Trend

S	Build ↑	Time Since	Duration
✓	#1	1 min 22 sec	3.5 sec

Icon: S M L

Builds >

Filter

Today

✓ #1 16:03

- **Check the status success 1st one.**

Jenkins / Github-01

- Status
- Changes
- Workspace
- Build Now
- Configure

✓ Github-01

Permalinks

- Last build (#1), 3 min 28 sec ago
- Last stable build (#1), 3 min 28 sec ago
- Last successful build (#1), 3 min 28 sec ago
- Last completed build (#1), 3 min 28 sec ago

- **Create 2nd GitHub -02 in new item**

New Item

Enter an item name

Github-02

Select an item type



Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

- **In Source Management change to Git and Paste URL**

Configure

- General
- Source Code Management**
- Triggers
- Environment
- Build Steps
- Post-build Actions

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ **Git** ?

Repositories ?

Repository URL ?

https://github.com/betawins/Techie_horizon_Login_app

- **In Build Steps add command and save it**

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

 **Execute shell** ?

Command

[See the list of available environment variables](#)

```
echo "Github-02 Job Success"
```

- **Successful output for 2nd Github-02**

First time build. Skipping changelog.

```
[Github-02] $ /bin/sh -xe /tmp/jenkins1092584683409072599.sh
+ echo 'Github-02 Job Success'
Github-02 Job Success
Finished: SUCCESS
```

- **Create New Item in Jenkins as GitHub -03**

New Item

Enter an item name

Github-03

Select an item type



Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications

- In Source Code change to git and paste Url

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ Git ?

Repositories ?

Repository URL ?

`https://github.com/betawins/Techie_horizon_Login_app`

- In Build steps change to Execute shell and given command and save

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

 Execute shell ?

Command

[See the list of available environment variables](#)

`echo "Github-03 Job Success"`

- Run the Build Now and check the status success

Build Time Trend

S	Build ↑	Time Since	Duration	
	#1	5.3 sec	0.56 sec	

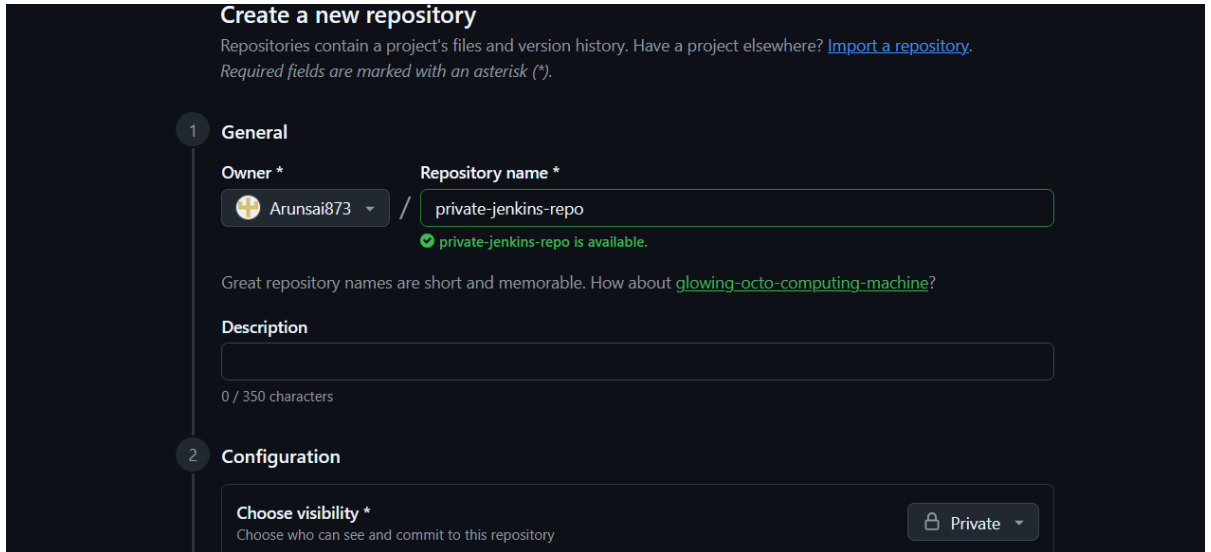
Icon: S M L

- Open console checks their Success

```
First time build. Skipping changelog.
[Github-03] $ /bin/sh -xe /tmp/jenkins3340733693322198804.sh
+ echo 'Github-03 Job Success'
Github-03 Job Success
Finished: SUCCESS
```

6. Create one Jenkins job using GitHub Private repository.

- Open GitHub and click on new repository and given name and in the configuration change private to public



Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1 General

Owner * Arunsai873 / Repository name *

private-jenkins-repo is available.

Great repository names are short and memorable. How about [glowing-octo-computing-machine?](#)

Description

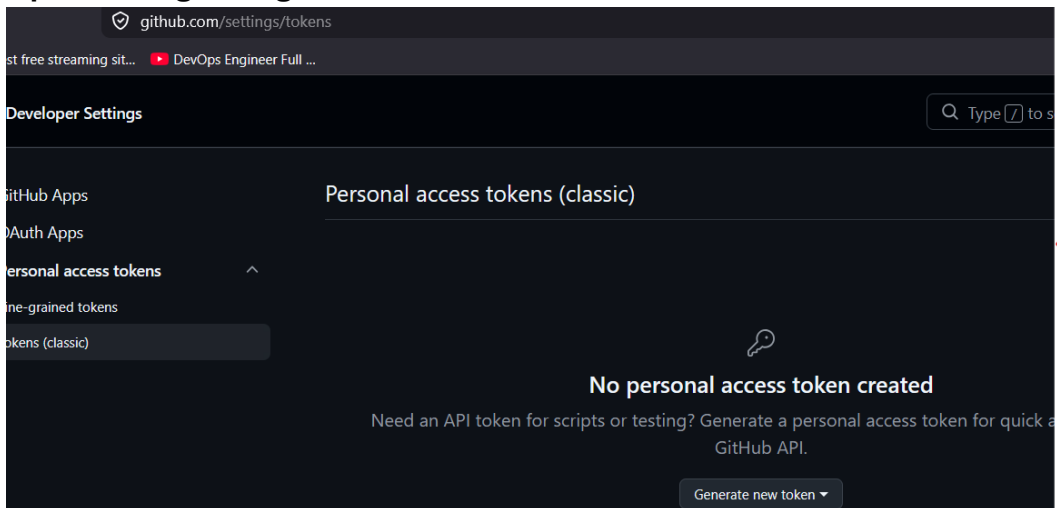
0 / 350 characters

2 Configuration

Choose visibility * Private

Choose who can see and commit to this repository

- Open settings and generate new token



github.com/settings/tokens

st free streaming sit... DevOps Engineer Full ...

Developer Settings

GitHub Apps

OAuth Apps

Personal access tokens

Fine-grained tokens

Tokens (classic)

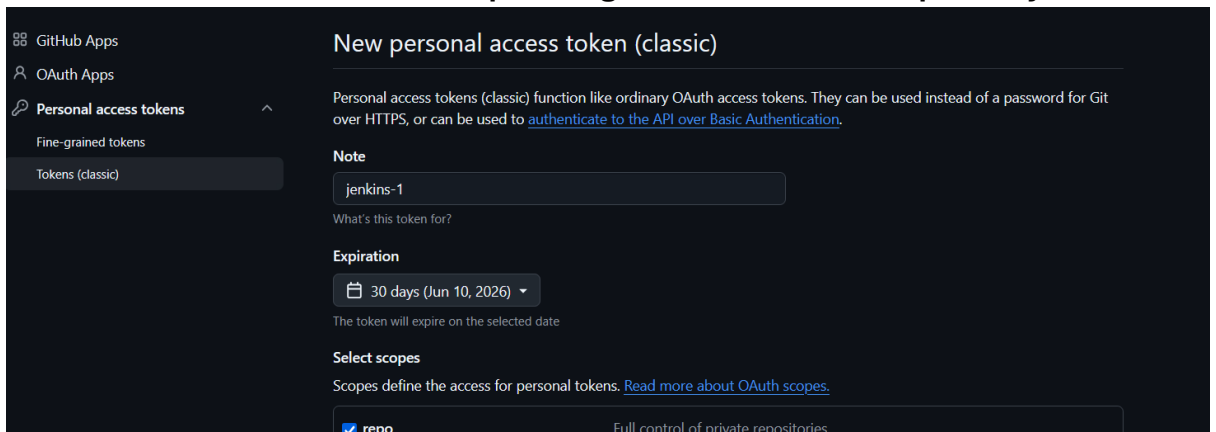
Personal access tokens (classic)

No personal access token created

Need an API token for scripts or testing? Generate a personal access token for quick access to the GitHub API.

Generate new token

- Given name in note and enable repo and generate token and kept safely.



GitHub Apps

OAuth Apps

Personal access tokens

Fine-grained tokens

Tokens (classic)

New personal access token (classic)

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to [authenticate to the API over Basic Authentication](#).

Note

jenkins-1

What's this token for?

Expiration

30 days (Jun 10, 2026)

The token will expire on the selected date

Select scopes

Scopes define the access for personal tokens. [Read more about OAuth scopes](#).

☒ repo Full control of private repositories

- **Now open Manage Jenkins in that Credentials**

Security

 **Security**
Secure Jenkins; define who is allowed to access/use the system.

 **Credentials**
Configure credentials

 **Credential Providers**
Configure the credential provide

 **Manage and Assign Roles**
Handle permissions by creating roles and assigning them to users/groups

 **Users**
Create/delete/modify users that can log in to this Jenkins.

- **Select Global and Click Add Credentials and select username and password**

Jenkins / Manage Jenkins / Credentials / System / Global

Global
Credentials that should be available everywhere.

Add Credentials

Select a type of credential

 **Username with password**
Commonly used for authentication to services like Git, APIs, or registries.

 **GitHub App**

 **SSH Username with private key**

 **Secret file**

- **Given details**

< Add Username with password X

Scope ?
Global (Jenkins, nodes, items, all child items, etc) v

Username
Arunasai873

☐ Treat username as secret ?

Password
.....

ID ?
private-jenkins-repo

Description ?
GitHub Private Repo

Create

- **Now created new item as GitHub Private**

/ New Item

New Item

Enter an item name

Private-GitHub-Job

Select an item type



Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

- **In Source Management change git and added my private git URL**

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ Git ?

Repositories ?

Repository URL ?

https://github.com/Arunsai873/private-jenkins-repo

Credentials ?

Arunsai873/***** (GitHub Private Repo)

+ Add

- **Now Build Steps and Execute Shell and command it and save**

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment

Execute shell ?

Command

See [the list of available environment variables](#)

```
echo "Private GitHub Job Success"
```

- **Build Now and check the console**

```
git checkout -b 1013311b20c74332200430001d7d1c100017c3b # checkout-10
Commit message: "Add files via upload"
First time build. Skipping changelog.
[Private-GitHub-Job] $ /bin/sh -xe /tmp/jenkins6137827825118450677.sh
+ echo 'Private GitHub Job Success'
Private GitHub Job Success
Finished: SUCCESS
```